Linux Fundamentals — Day 2

Intermediate Linux & Administration · Study Notes

Topics: user management, packages, process monitoring, networking, system administration, and troubleshooting.

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1. User Management and Package Installation

1.1 Linux user accounts

Linux supports multiple users and services, each operating with an identity and permissions. Carefully managing accounts helps reduce unnecessary access.

Term	Meaning
UID	Numeric user identifier; root normally has UID 0
GID	Numeric group identifier
Home directory	Default location for a user's files, often `/home/username`
Shell	Command interpreter assigned to a user
`/etc/passwd`	Account metadata such as username, UID, GID, home directory, and shell
`/etc/shadow`	Password hashes and password-aging information; access is restricted
`/etc/group`	Group definitions and membership information

1.2 Create and manage users

`adduser` is an interactive, high-level helper on many Debian-based systems. `useradd` is a lower-level command and often requires options to create a home directory and select a shell.

```
sudo adduser alice
sudo useradd -m -s /bin/bash bob
sudo useradd -m -G sudo carol
sudo passwd alice
```

Modify accounts and groups:

```
sudo usermod -aG sudo alice       # Add alice to sudo group
sudo usermod -aG docker,www-data bob
sudo usermod -L alice             # Lock account
sudo usermod -U alice             # Unlock account
sudo userdel alice                # Remove account; home normally remains
sudo userdel -r alice             # Remove account and home directory
```

Be careful with account deletion: `userdel -r` removes the user's home directory and may delete personal data. Confirm the target account and backups first.

1.3 Password basics

```
passwd                    # Change your own password
sudo passwd alice         # Set/change another user's password
sudo passwd -l alice      # Lock password
sudo passwd -u alice      # Unlock password
```

Use strong, unique passwords and follow the system's account-management policy.

1.4 Package management with APT

`apt` is commonly used on Debian-based distributions, including Ubuntu and Kali Linux, to search, install, upgrade, and remove software.

```
sudo apt update              # Refresh package metadata
sudo apt upgrade             # Upgrade installed packages
sudo apt install nmap        # Install a package
sudo apt install curl wget   # Install multiple packages
sudo apt remove nmap         # Remove a package
sudo apt autoremove          # Remove unused dependencies
sudo apt search nmap         # Search package metadata
apt show nmap                # Display package details
```

Tool	Package name	Typical purpose
Nmap	`nmap`	Network discovery and port scanning
curl	`curl`	Transfer data to/from URLs and test APIs
wget	`wget`	Download files
Netcat	`netcat-openbsd`	Network troubleshooting and connection testing
Hydra	`hydra`	Password-auditing tool; use only with explicit authorization
tree	`tree`	Display directory structures
John the Ripper	`john`	Password-hash auditing in authorized assessments

Lab — Users and packages

1. Refresh package metadata with `sudo apt update`.
2. Install `tree`, then run `tree /home`.
3. Create a test account with `sudo adduser testuser`.
4. Verify the account using `grep testuser /etc/passwd`.
5. Change its password using `sudo passwd testuser`.
6. Inspect its groups with `groups testuser`.
7. Install Nmap and check its version with `nmap --version`.

Assessment questions

1. What is the difference between `adduser` and `useradd`?
2. What does `apt update` do?
3. How can an existing user be added to the `sudo` group?
4. What normally happens to a home directory when `userdel` is used without `-r`?
5. How can an account be locked without deleting it?

2. Process and System Monitoring

2.1 What is a process?

A process is a running instance of a program or command. Each process has a Process ID (PID). Processes may launch other processes and can run in the foreground, background, or as services.

Term	Meaning
PID	Process ID
PPID	Parent Process ID
Foreground	Runs interactively in the terminal

Term	Meaning
Background	Runs without directly occupying the terminal, often started with &
Daemon/service	Long-running background process, such as an SSH service

2.2 Foreground and background processes

```
sleep 300 &                # Start sleep in the background
jobs                       # Show jobs in the current shell
```

2.3 `ps` — process snapshot

```
ps                         # Show processes associated with the current terminal
ps aux                     # Show processes from all users
ps aux | grep ssh          # Filter output for SSH-related processes
pgrep apache2              # Find process IDs matching a name
```

Processes running from unusual locations, consuming unexpected resources, or using unfamiliar names may deserve investigation. A process located in `/tmp` is not automatically malicious; examine its owner, executable, command line, and context.

2.4 `top` — real-time monitoring

```
top
```

Useful keys inside `top` include:

- q: quit
- `M`: sort by memory usage
- `P`: sort by CPU usage
- k: request termination of a process by PID

2.5 Terminate processes

```
kill 1247                  # Send SIGTERM by default
kill -9 1247               # Send SIGKILL; forceful, last resort
pkill apache2              # Signal processes matching a name
```

Prefer normal termination first. `SIGTERM` gives a process a chance to clean up; `SIGKILL` cannot be caught and may result in lost work. Verify the PID before sending a signal.

2.6 Command history and security

```
history                    # Show shell history
history | tail -20         # Show recent commands
history | grep ssh         # Search history for SSH-related commands
```

Shell history can unintentionally contain sensitive data if passwords, tokens, or keys are typed directly into commands. Avoid placing secrets in command arguments. History files and logging behavior vary by shell and configuration.

Lab — Process monitoring

1. Run `ps aux` and identify three processes and their PIDs.
2. Use `pgrep ssh` or `ps aux | grep ssh` to inspect SSH-related processes.
3. Run `top`, observe CPU and memory usage, then press `q`.
4. Run `history | tail -20`.
5. Start `sleep 300 &`, record its PID, and stop it with `kill <PID>`.
6. Verify that the process has stopped.

Assessment questions

1. What is a PID, and which command lists processes across users?
2. How do `SIGTERM` and `SIGKILL` differ?
3. What might make a process suspicious during an investigation?

4. Why can shell history matter in a security investigation?
5. What does appending & to a command do?

3. Networking Basics in Linux

3.1 View network interfaces with `ip`

The `ip` command is the modern standard for inspecting network interfaces and addresses.

```
ip a                      # Show interfaces and IP addresses
ip a show eth0             # Show a particular interface
ip route                   # Show routes and the default gateway
```

The loopback address `127.0.0.1` refers to the local machine. Private addresses such as `192.168.1.45` are commonly used on local networks; actual addresses depend on your network.

3.2 Legacy tools: `ifconfig` and `iwconfig`

`ifconfig` is an older interface tool. `iwconfig` provides information about wireless interfaces on systems where the legacy wireless tools are installed. Modern systems may use other wireless utilities.

```
ifconfig
ifconfig eth0
iwconfig
iwconfig wlan0
sudo apt install net-tools wireless-tools
```

Command	Typical purpose
`ip a`	Show IP addresses and interfaces
`ifconfig`	Legacy interface inspection/configuration
`iwconfig`	Legacy wireless interface information
`ip route`	View routing table and default route

3.3 Test connectivity with `ping`

```
ping google.com
ping -c 4 google.com
ping -c 1 192.168.1.1
```

Some networks block ICMP, so a failed ping does not always mean the host is offline.

A basic ping sweep can send traffic to many addresses. Only perform discovery on networks you own or are explicitly authorized to test. Prefer a controlled lab and a narrow, approved range.

3.4 Hostname and local addresses

```
hostname
hostname -I
sudo hostnamectl set-hostname newname
```

Changing the hostname affects system configuration; use an appropriate name and administrative authorization.

3.5 SSH — Secure Shell

SSH provides encrypted remote terminal access and can also run a remote command.

```
ssh username@192.168.1.100
ssh -p 2222 username@192.168.1.100
ssh -v username@192.168.1.100
ssh -i ~/.ssh/id_ed25519 username@192.168.1.100
ssh username@192.168.1.100 'cat /etc/passwd'
```

Replace example usernames, addresses, ports, and paths with those for your authorized environment.

3.6 Generate SSH keys

SSH key pairs contain a private key and a public key. Keep the private key secret; install or share the public key according to the server's access policy.

```
ssh-keygen -t ed25519 -C 'user@lab'
```

```
ssh-keygen -t rsa -b 4096 -C 'user@lab'
cat ~/.ssh/id_ed25519.pub
ssh-copy-id username@192.168.1.100
```

Protect private keys with appropriate filesystem permissions and a passphrase when suitable.

Algorithm	Notes
Ed25519	Modern, efficient option supported by current OpenSSH deployments
RSA	Widely compatible; use an appropriate key size and modern configuration
ECDSA	Supported by many systems; suitability depends on policy and implementation
DSA	Obsolete for modern SSH use; avoid

Lab — Networking and SSH

1. Run `ip a` and identify your active interface and IP address.
2. Test connectivity with `ping -c 4 8.8.8.8`.
3. Run `hostname` and `hostname -I`.
4. Generate a lab SSH key with `ssh-keygen -t ed25519 -C 'mylab'`.
5. Display the public key using `cat ~/.ssh/id_ed25519.pub`.
6. Do not share the private key.

Assessment questions

1. How does `ip a` differ from `ifconfig`?
2. What is the difference between a public and private SSH key?
3. What SSH command connects to a host on a non-standard port?
4. What is the purpose of SSH verbose mode (`-v`)?
5. Why must private keys be protected?

4. System Administration and Troubleshooting

4.1 Disk space: `df` and `du`

`df` reports space usage for mounted filesystems. `du` estimates space used by files and directories.

```
df -h                       # Filesystem usage in readable units
df -h /home                 # Check the filesystem containing /home
df -T                       # Include filesystem type
du -h /home                 # Directory usage
du -sh /home                # Summary total for /home
du -sh /* 2>/dev/null | sort -rh | head -10
du -h --max-depth=1 /var    # Summarize immediate subdirectories
```

4.2 Uptime and CPU count

```
uptime
nproc
```

`uptime` reports how long the system has been running, logged-in users, and load averages over 1, 5, and 15 minutes. Interpret load averages in context: CPU count, workload type, and I/O wait all matter.

4.3 Update the system

```
sudo apt update
sudo apt upgrade
sudo apt dist-upgrade
sudo apt autoremove
```

Review proposed changes before confirming upgrades, especially on production systems. `dist-upgrade` may install or remove packages to resolve dependencies.

4.4 Common errors

Error	Possible cause	First checks
`Permission denied`	Insufficient permissions or restrictive file mode	`ls -l`; confirm correct account and access policy
`Command not found`	Program missing or not in `PATH`	Check spelling, `command -v name`, package availability
`No space left on device`	Filesystem or inode exhaustion	`df -h`; consider `df -i`
`Network is unreachable`	Missing route or network configuration issue	`ip a`, `ip route`, connectivity tests
`Connection refused`	Service not listening or wrong port	Check service status and `ss -tuln`
`Too many open files`	File-descriptor limit reached	`ulimit -n`; inspect application/service limits

Use `sudo` only when necessary and authorized; it is not a universal fix for errors.

4.5 Quick system health check

Run these commands on a system you administer and review the output:

```
whoami && id
df -h
uptime
ps aux | head -20
ss -tulnp
last | head -10
```

On some distributions, failed-login logs may be stored in a different file or system journal. On systems where `/var/log/auth.log` exists:

```
sudo grep 'Failed' /var/log/auth.log | tail -10
```

To locate world-writable regular files, a search may be useful but can take time and produce permission errors:

```
find / -type f -perm -o+w 2>/dev/null | grep -v '^/proc/'
```

Review findings in context rather than assuming every match is a vulnerability.

Lab — System administration

1. Run `df -h` and identify the filesystem with the highest usage.
2. Run `du -sh /var/log` to estimate log-directory size.
3. Run `uptime` and record uptime and load averages.
4. Update package metadata and review available upgrades.
5. Run `ss -tulnp` and identify listening services.
6. Run `who` to see logged-in users.
7. Save selected command output to a file, for example: `df -h > disk-report.txt`.

Assessment questions

1. How do `df` and `du` differ?
2. How should load average be interpreted relative to CPU count and workload?
3. What are two safe steps to investigate `Permission denied`?
4. Which command displays listening sockets?
5. What information is useful in a basic system health check?

5. Learning Outcomes and Final Practical

After completing these topics, you should be able to:

- Explain Linux user and group concepts.
- Create and manage user accounts carefully.

- Install and maintain packages using APT.
- Inspect processes and system resource usage.
- Review basic network configuration and test connectivity.
- Understand SSH access and key-pair handling.
- Check disk usage, uptime, listening ports, and common errors.

Final practical checklist

- Record `whoami` and `id`.
- Check storage with `df -h`.
- Record system uptime using `uptime`.
- Inspect running processes using `ps aux`.
- Review listening sockets using `ss -tulnp`.
- Check network interfaces with `ip a`.
- Test connectivity with an approved target.
- Document any errors, evidence, and corrective actions.

Safety reminder: Run administrative and security commands only on systems you own or are authorized to manage. Avoid sharing passwords, tokens, or private keys in terminal output or repositories.